

Math 105LA Assignment 7

Fall 2019

Due Sun 12/01, 11pm

Write a MATLAB function to implement the SOR iteration to solve the linear system $Ax = b$.

The algorithm for SOR iteration is

Input: a matrix, A , a vector b , weight w , initial guess x_0 , tolerance tol , tolerance for the residual tol_r , maximum number of iteration N
Output: the approximate solution x
Step 1: Set $k = 1$;
Set $n =$ number of rows of A .
Step 2: While $k \leq N$, repeat step 3 to step 6
Step 3: For $i = 1, 2, \dots, n$
Set $x_i = (1 - w) * x_{0i} + w * (-\sum_{j=1}^{i-1} (a_{ij}x_j) - \sum_{j=i+1}^n (a_{ij}x_{0j}) + b_i) / a_{ii}$
Step 4: If $\|Ax - b\| < tol_r$
Output('||Ax-b|| is smaller than tol_r', x), stop
Step 5: If $\|x - x_0\| < tol$
Output('||x_{n+1} - x_n|| is smaller than tol', x), stop
Step 6: Set $x_0 = x$;
Set $k = k + 1$
Step 7: Output('Maximum number of iterations exceeded', x), stop

You can start with the template below. Make sure that you use the **same order** of input:

```
function x = SOR(A,b,w,x0,N,tol,tol_r)
% A,b = the linear system A*x=b to be solved
% x0 = initial guess
% tol = tolerance
% tol_r = tolerance for the residual
% N = maximum number of iterations

end
```

For example, suppose that we would like to solve linear system

$$\begin{cases} 2x_1 - x_2 + x_3 = 1 \\ -x_1 + 2x_2 - 2x_3 = 4 \\ x_1 - 2x_2 + 3x_3 = -5 \end{cases}$$

within 10^{-6} accuracy, using $w=3/2$, initial guess $x_0 = [0, 0, 0]^T$ and at most 300 iterations. First define the coefficient matrix, right-hand side vector and the initial guess:

```
A = [2 -1 1; -1 2 -2; 1 -2 3];
b = [1; 4; -5];
w = 3/2;
x0 = [0; 0; 0];
```

Then, assuming your function is saved in 'SOR.m', type the following in the command window:

```
x = SOR(A,b,w,x0,300,1e-6,1e-6)
```

And your function should return

$||x_{n+1}-x_n||$ is smaller than $1.00e-06$.

```
x =
    2.0000
    2.0000
   -1.0000
```

More examples:

Input	And it should return ...
<pre>A = [2 -1 1; -1 2 -2; 1 -2 3]; w = 1; b = [-1; 4; -5]; x0 = [0; 0; 0]; SOR(A,b,w,x0,300,1e-6,1e-6);</pre>	<pre> Ax-b is smaller than 1.00e-06. ans = 0.6667 1.3333 -1.0000</pre>
<pre>A = [10 5 0 0; 5 10 -4 0; 0 -4 8 -1; 0 0 -1 5]; b = [6; 25; -11; 11]; x0 = [0; 0; 0; 0]; w = 1; x = SOR(A,b,w,x0,10,1e-6,1e-6) error = norm(x-A\b) / norm(x0-A\b) w = 1.999; x = SOR(A,b,w,x0,10,1e-6,1e-6) error = norm(x-A\b) / norm(x0-A\b) w = 2/(1+sqrt((1-0.6793055^2))); x = SOR(A,b,w,x0,10,1e-6,1e-6) error = norm(x-A\b) / norm(x0-A\b)</pre>	<pre>Maximum number of iterations exceeded. x = -1.0034 3.2082 0.5170 2.3034 error = 4.4977e-04 Maximum number of iterations exceeded. x = -3.7993 5.5227 0.8132 4.8063 error = 1.0750 Maximum number of iterations exceeded. x = -1.0047 3.2094 0.5176 2.3035 error = 1.8288e-07</pre>

Submit a single file **named by your UCI netID** (e.g. wtleung.m) that contains the function. Your code will be tested on a few problems. Your grade of this assignment is calculated by the number of problems that your code solves correctly. Make sure your function runs without errors and has **the same order of input** above. **Failure to do so will result in a zero grade.**